

A photograph of a soldier in camouflage gear using a ruggedized laptop in a field setting. The soldier's hands are visible, and the laptop is open, showing a dark screen. The background is blurred, showing more of the soldier's gear and the field environment. The image is framed by a dark semi-transparent rectangle.

MIL-STD-1472G: Graphical User Interfaces (GUIs)

Steph Schuman

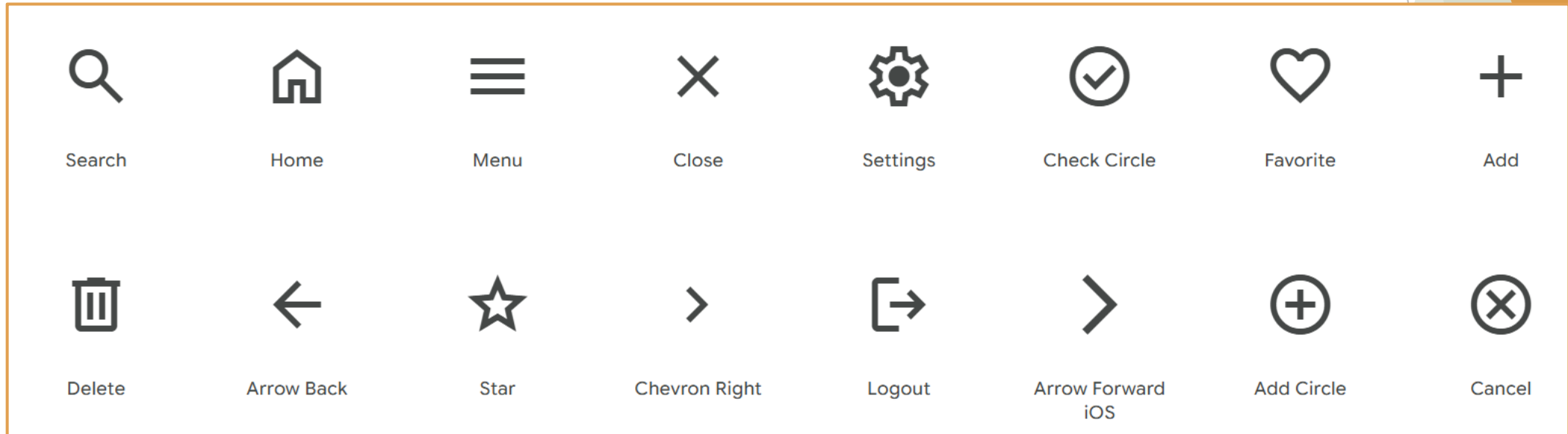
MIL-STD-1472G

- ▶ Department of Defense Design Criteria Standard
- ▶ General human engineering criteria
- ▶ Design and development of military:
 - ▶ Systems
 - ▶ Equipment
 - ▶ Facilities
- ▶ Topics:
 - ▶ Controls
 - ▶ Displays
 - ▶ Automated systems (UAV, UGV, UUV, USV)



Symbol Coding (5.2.2.3.8)

- ▶ Enhance information assimilation
- ▶ Analogs of the event or element they represent
- ▶ Symbol in general use and well known to the expected users
 - ▶ Population stereotypes
 - ▶ Material Design Icons



Fonts.google.com/icons


Fonts


Noto


Icons


Knowledge


FAQ

 Reset all 

 Customize

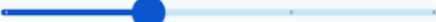
Fill  

Weight 

100 700

Grade 

-25 (low) 200 (high emphasis)

Optical Size 

20px 48px

Filter

 Style 

Material Symbols (new) 

 Google Fonts

 Search icons

Sort by
Most popular 



 Filters

Material Design icon guidelines
Learn about the latest best practices for icons 

Figma plugin
Use the Material Symbols plugin on Figma 

GitHub repo
This repository contains the binary font file served by Google Fonts 

Apache license
Material Symbols are available under the Apache License Version 2.0 

UI actions

 Search

 Home

 Menu

 Close

 Settings

 Check Circle

 Favorite

 Add

 Delete

 Arrow Back

 Star

 Chevron Right

 Logout

 Arrow Forward iOS

 Add Circle

 Cancel

 Arrow Back iOS

 Arrow Forward

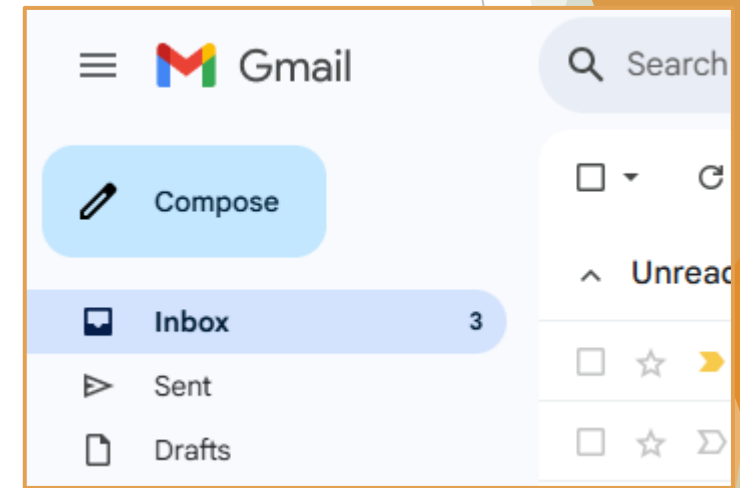
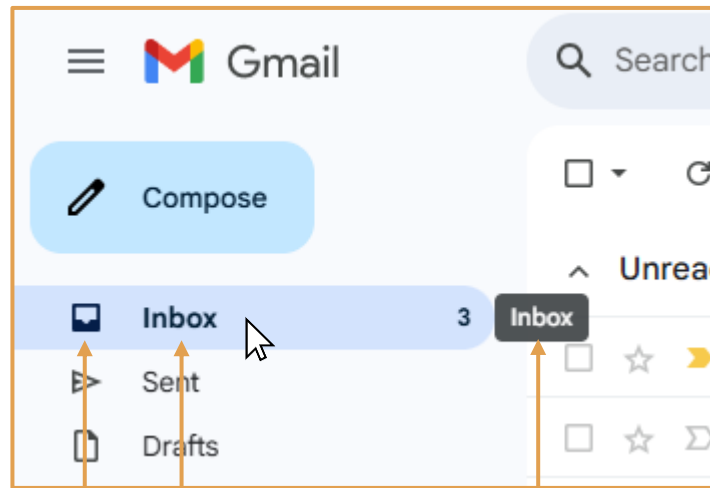
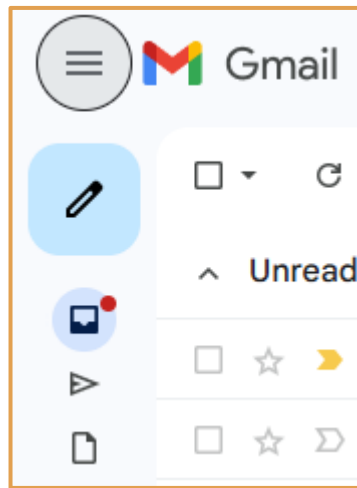
 Arrow Drop

 More Vert

 Check

Symbol Coding (5.1.3.5.7c.)

- ▶ Supplementary labels available
 - ▶ Ensure intended meaning



Color-Coding (5.2.2.3.2)

- Differentiate between information in dense displays
- ≤ 7 colors for color recognition

SalesOrderID ▼	RevisionNumber ▼	OrderDate ▼	ProductName ▼
43660	8	5/31/2011	Road-650 Red, 44
43660	8	5/31/2011	Road-450 Red, 52
43663	8	5/31/2011	Road-650 Red, 60
43666	8	5/31/2011	Road-650 Red, 52
43666	8	5/31/2011	Road-150 Red, 56



Color Contrast Checker

Calculate the contrast ratio of text and background colors.

Text color

#112A46



Background color

#ACC8E5



Contrast

8.42

Very good



Small text



Large text



Good contrast for small text (below 18pt) and great contrast for large text (above 18pt or bold above 14pt). [Click to enhance](#)

Quote n. 16

Even if you're on the right track, you'll get run over if you just sit there.

Will Rogers



Squarespace tools make it easy to create a beautiful and unique website.

ADS VIA CARBON

HIDE

Color-Coding (5.2.2.3.2)

- ▶ Colors have meaning
 - ▶ Prevent mismatch
- ▶ Warm - action, response requirement
- ▶ Cool - infrequent, background information

Red - danger, severe threat

Orange - high threat

Yellow - warning, impending danger

Green - safe, low threat



Blue - safe

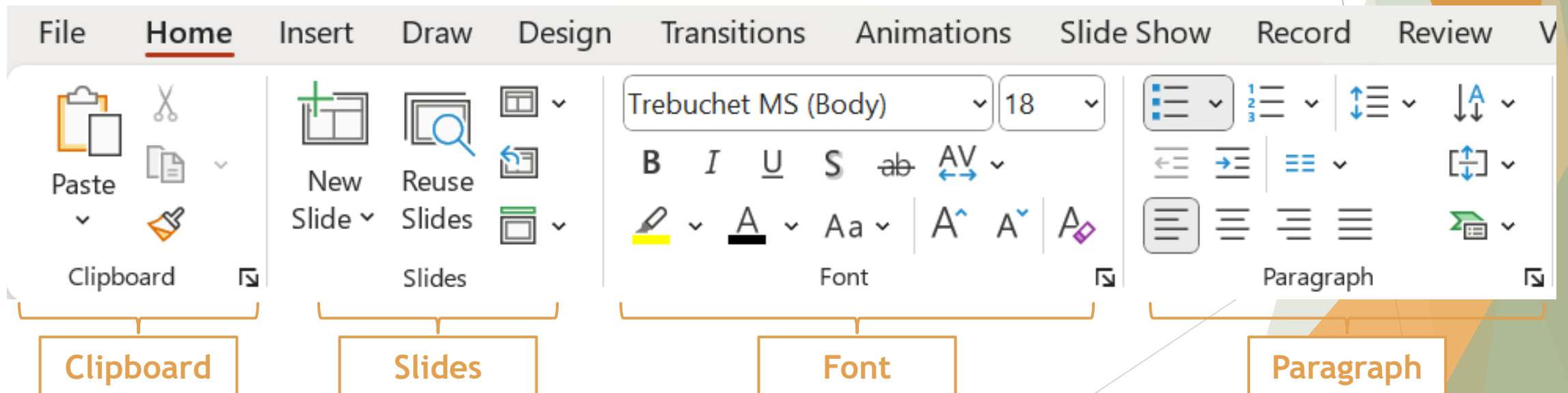
Cyan - advisory

Dark blue - advisory

Magenta - alarm, radiation

Functional Grouping (5.1.2.2.1)

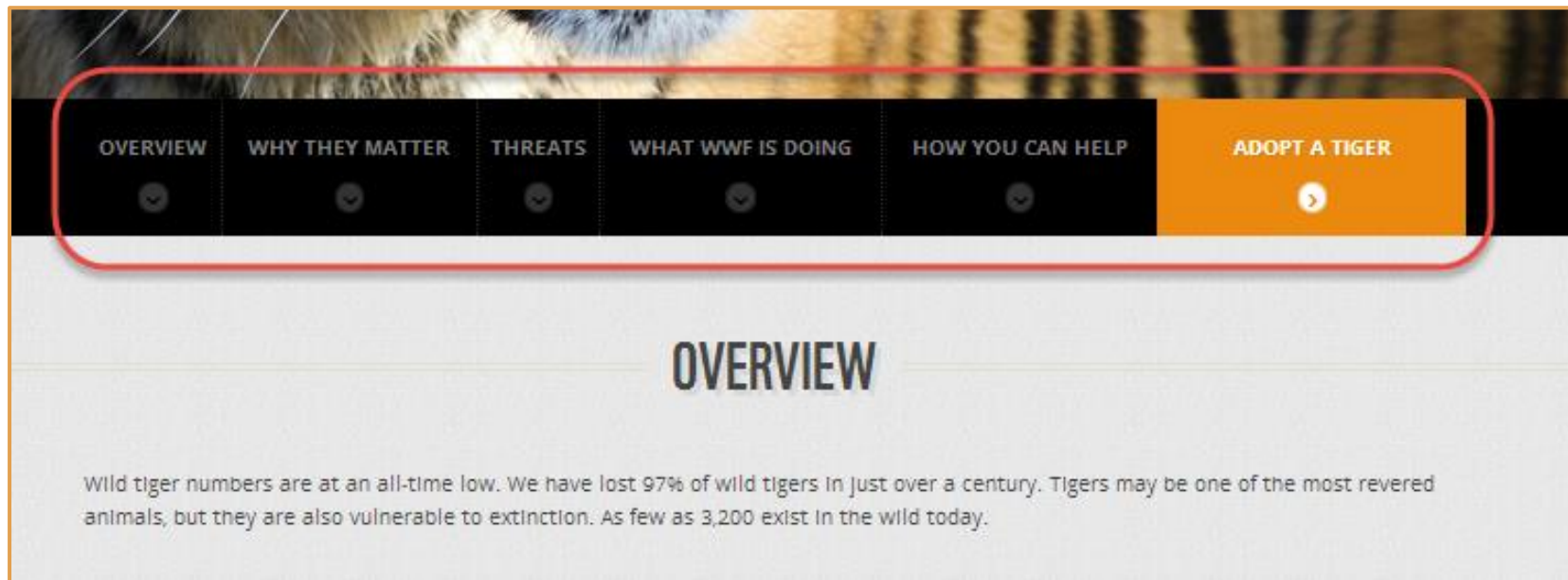
- ▶ Close to each other and in groups
- ▶ Easy access to frequently used controls



Functional Grouping (5.1.2.2.1)

- Sequential controls
 - Left-to-right or top-to-bottom

Title	Mr
First name	Graham
Last name	Charlton
Phone number	02072691450



Quantitative Information (5.2.2.2.3c)

- ▶ Numeric digital displays for precision
- ▶ Units of measure
- ▶ Just because you have a level of precisions doesn't mean you should display that many
- ▶ # decimals depends on what is actually relevant
 - ▶ i.e. lat lon needs a lot but alt doesn't

	Command					Lat	Long	Alt	Delete	Up	Down	Grad %	Dist	AZ
1	WAYPOINT ▼	0	0	0	0	-35.0407928	117.8277898	100	X	⬆	⬇	95.7	104.5	1
2	WAYPOINT ▼	0	0	0	0	-35.0406786	117.8260410	100	X	⬆	⬇	0.0	159.7	275
3	WAYPOINT ▼	0	0	0	0	-35.0417239	117.8251612	100	X	⬆	⬇	0.0	141.2	215

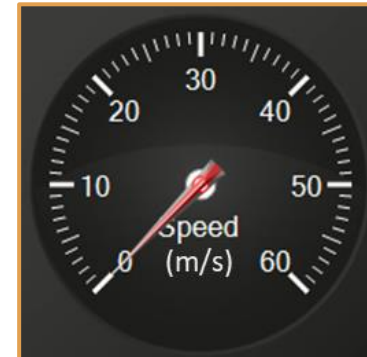
Spatial Display (5.12.3.1.14-15)

- ▶ Quantitative precision not important
- ▶ Minimize spatial disorientation
- ▶ Quick and accurate shifting frames of reference



Scales (5.2.2.5.3b)

- ▶ Quantitative and qualitative (direction of motion)
- ▶ Increase clockwise, left to right, or bottom up
- ▶ Increment by decimal multiples 1, 2, 5 units
- ▶ Moving pointer, fixed scale preferred over fixed pointer, moving scales
 - ▶ Unless length precludes fixed-scale format
- ▶ Circular scale preferred for speed
- ▶ Units



Thank you!

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